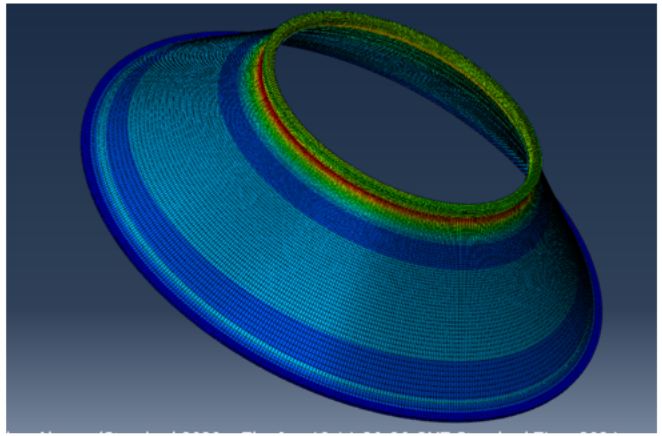
Projects

- SHIVA RAMKUMAR

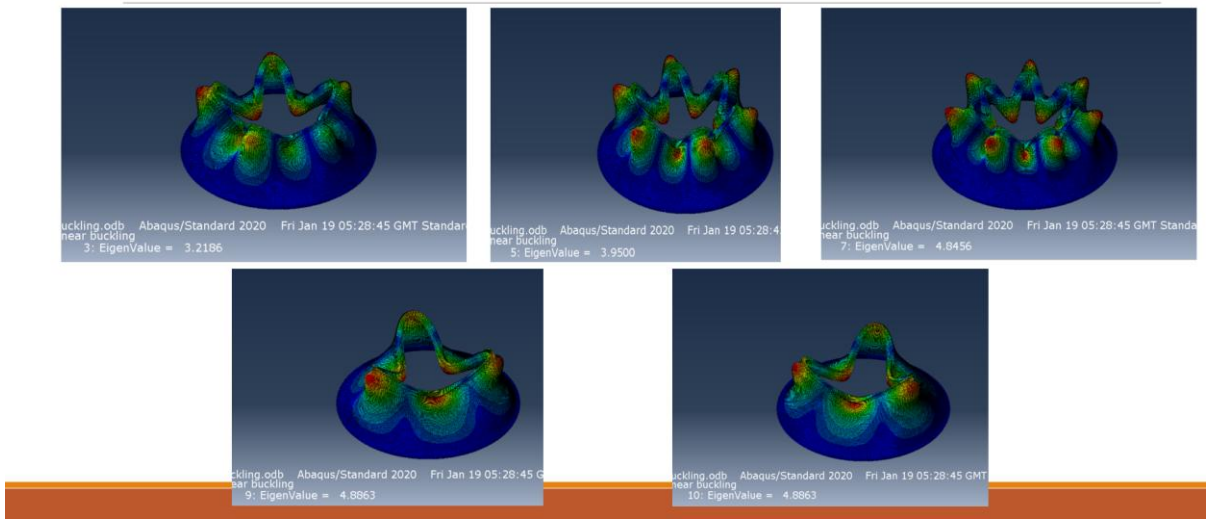
FINITE ELEMENT ANALYSIS

Payload Adapter System

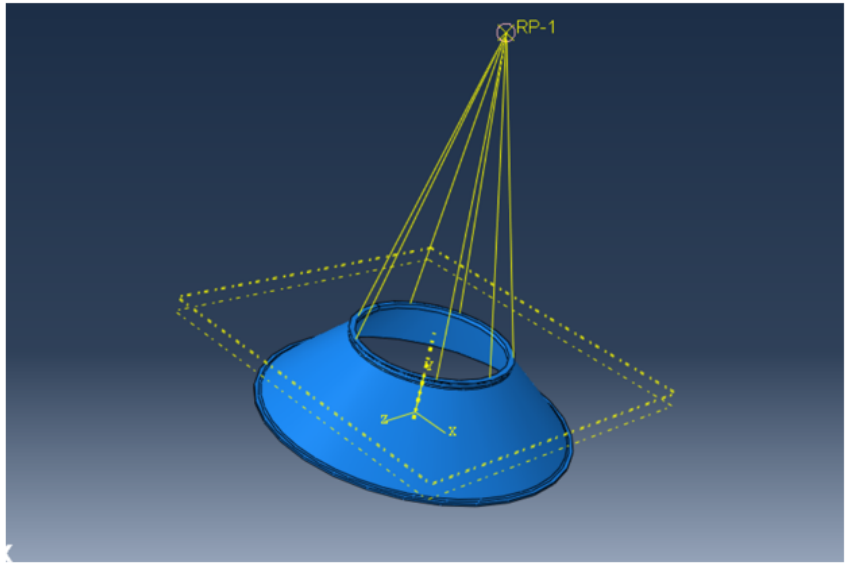
(PAS 937) FOR SOYUZ LAUNCHER



Linear Buckling Analysis



Load applied as Concentrated Force using a reference point

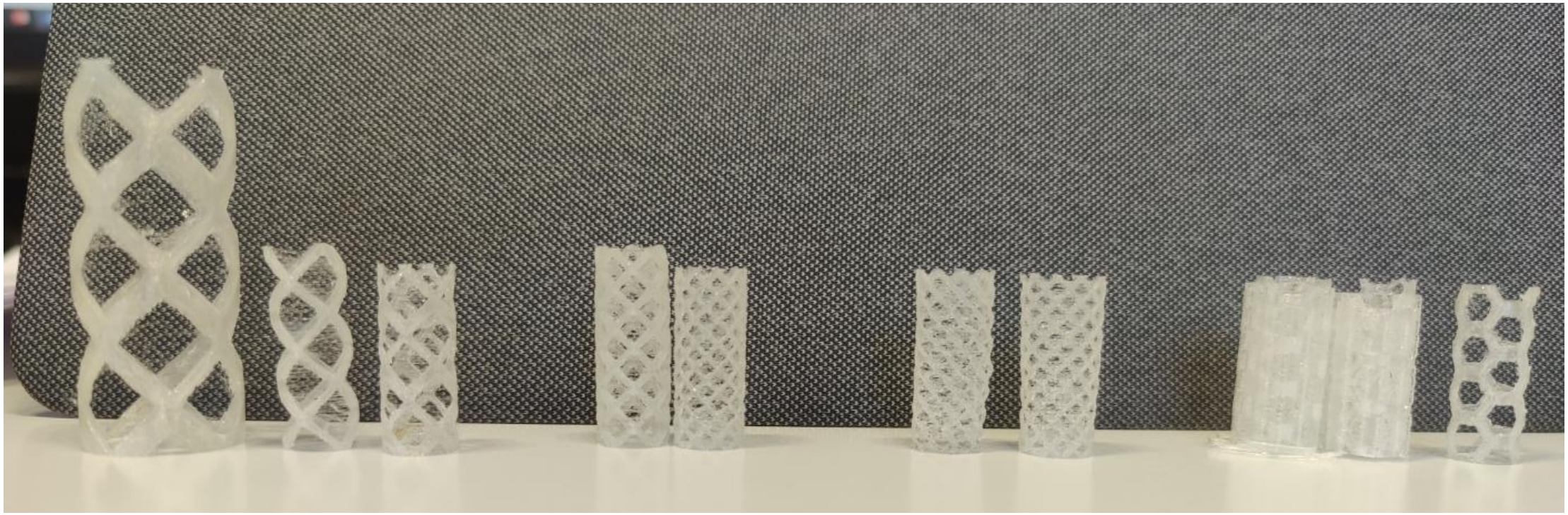


The PAS 937 S is designed and qualified to support a payload of 3500 kg centered at 1.8 m from the separation plane. (from the Soyuz manual)

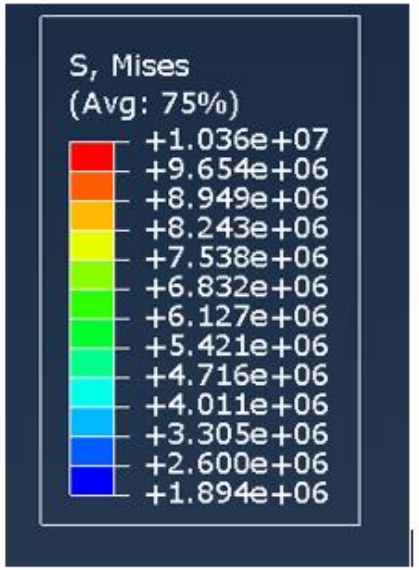
Load in axial direction under maximum acceleration (3500*9.81*4.3)/0.00974 = 1515816.2217659

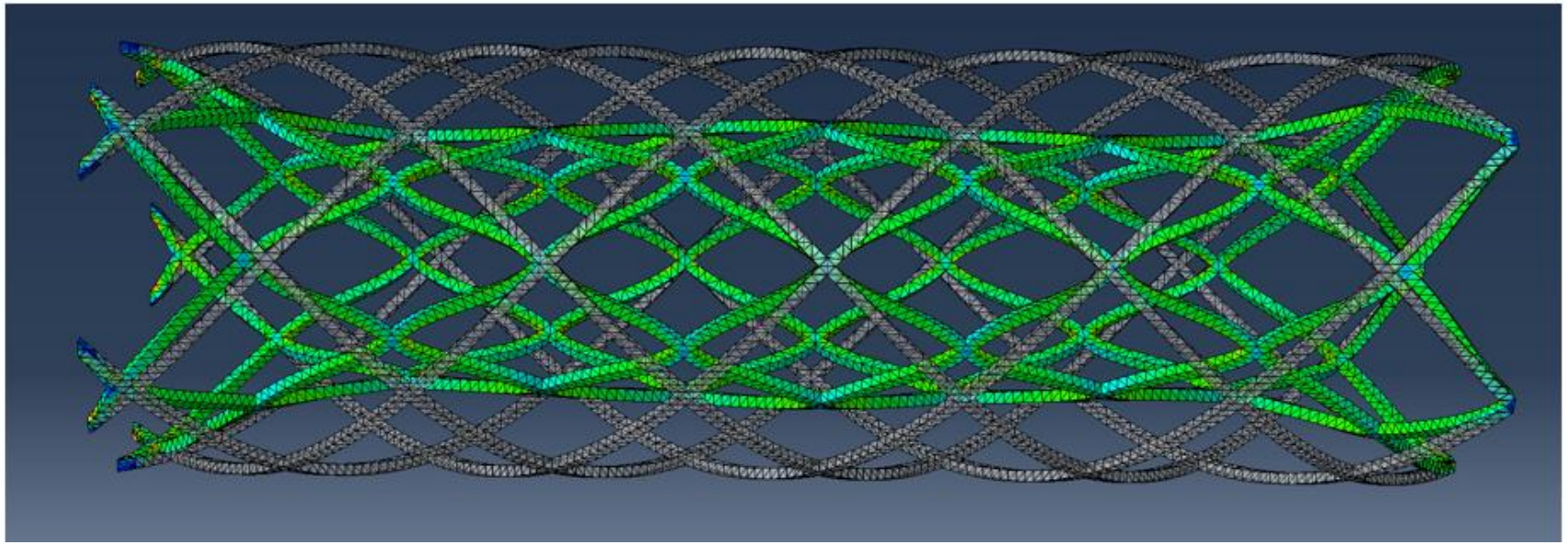
Shear force experienced when the Rocket is travelling at an angle of 45 degrees 0.4g =13734

4D printing of Vascular stents using PLA

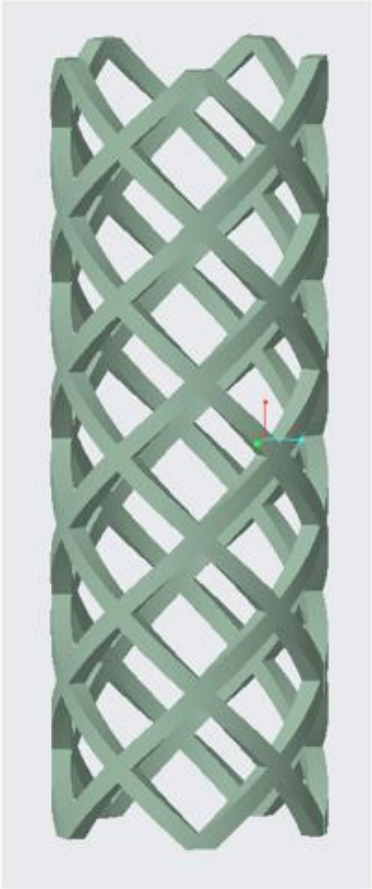


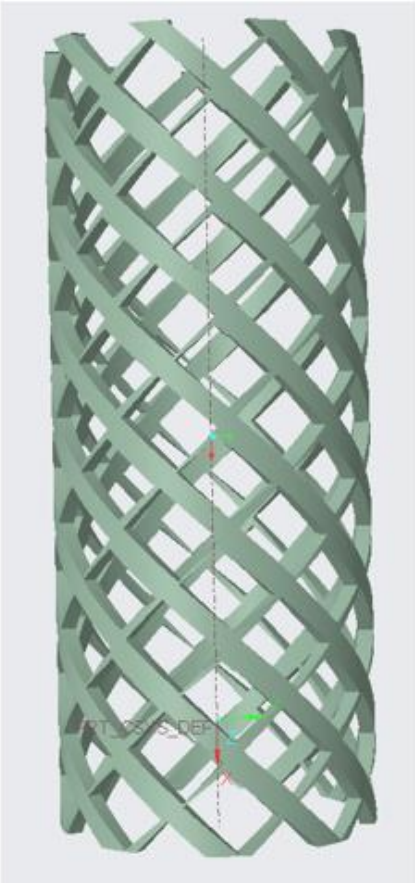
FEA ANALYSIS

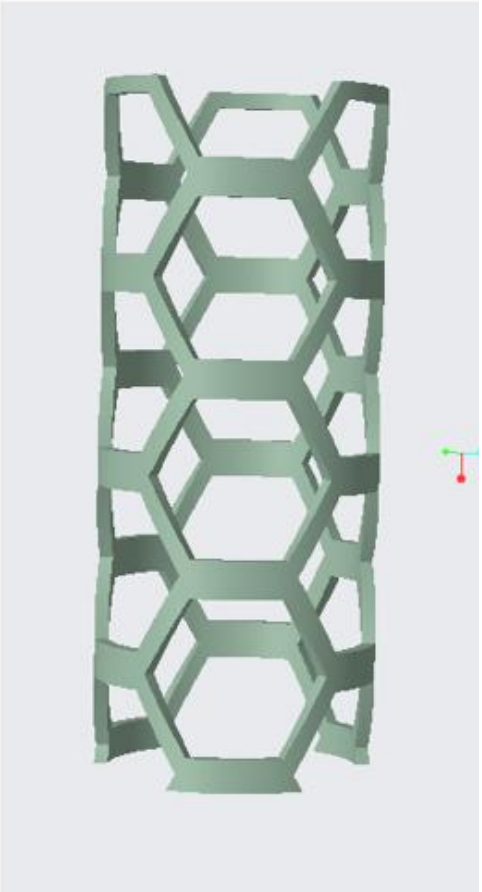


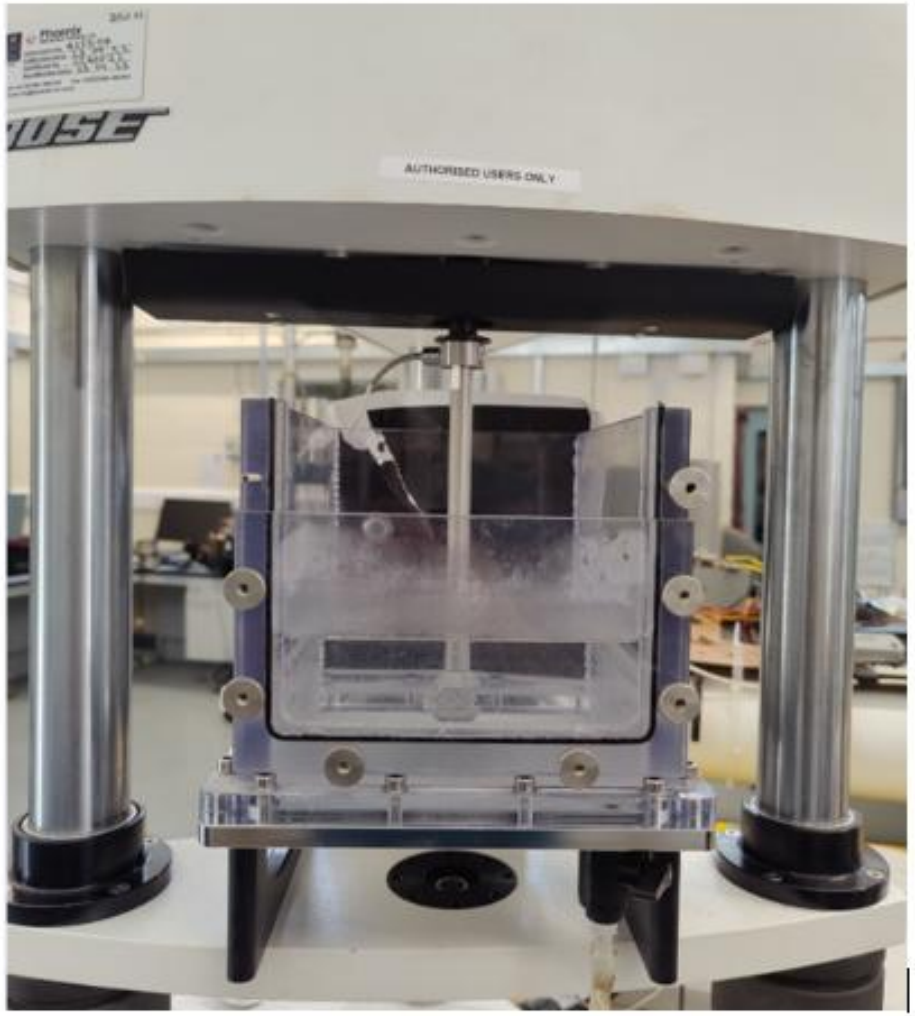


Designs using CAD for expansion upon external stimulus on 4D Material

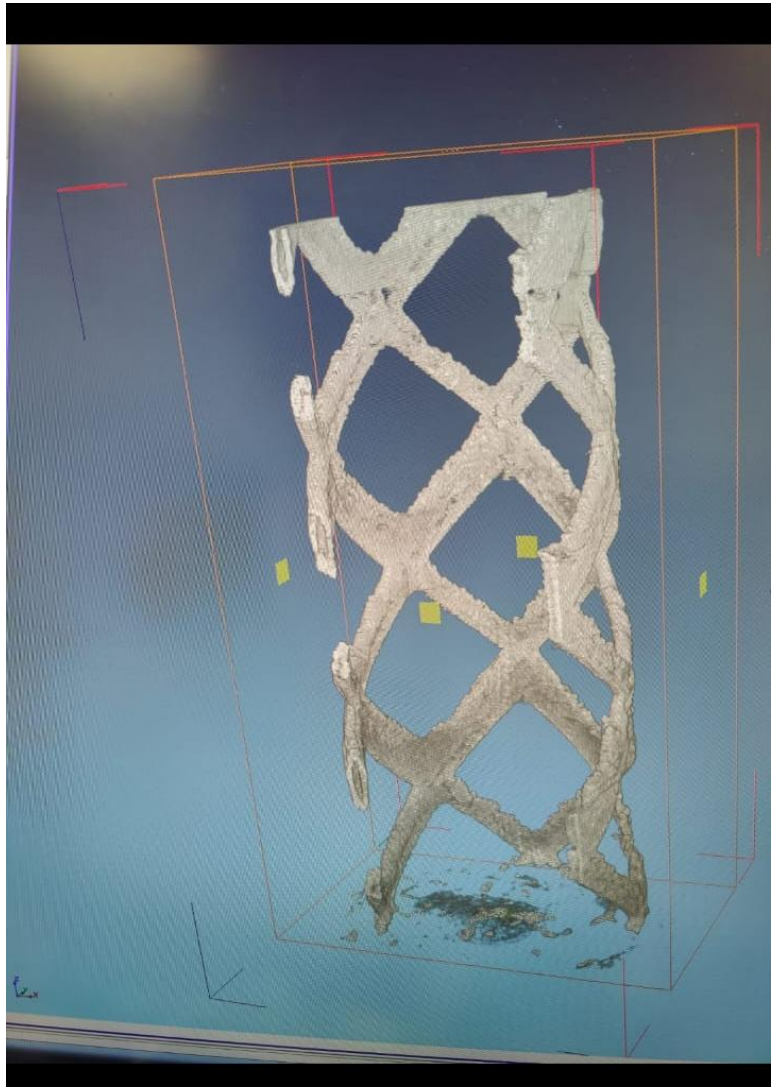








Compression Test

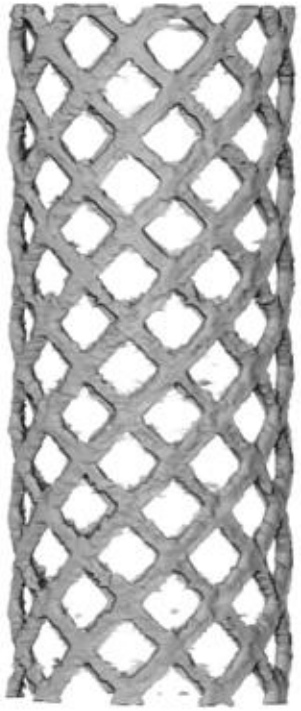


Porosity Test

CT Scans

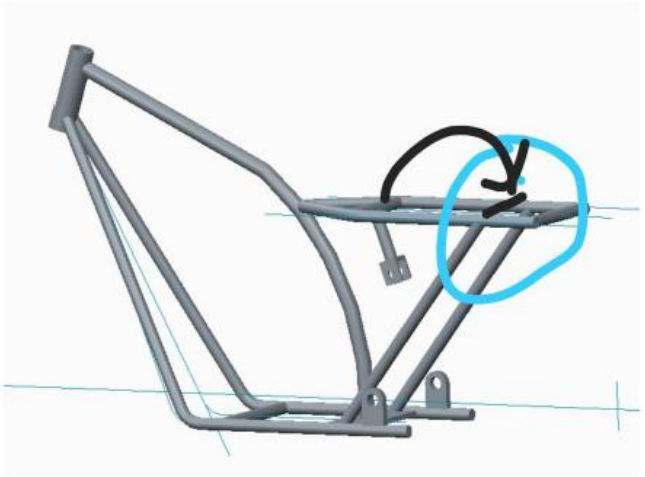
Stent A

Stent B

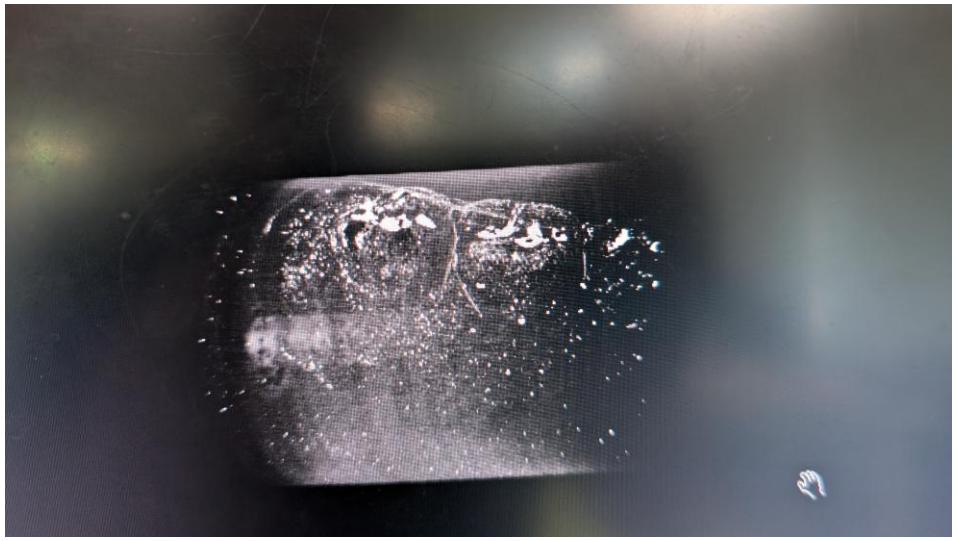


Stent C

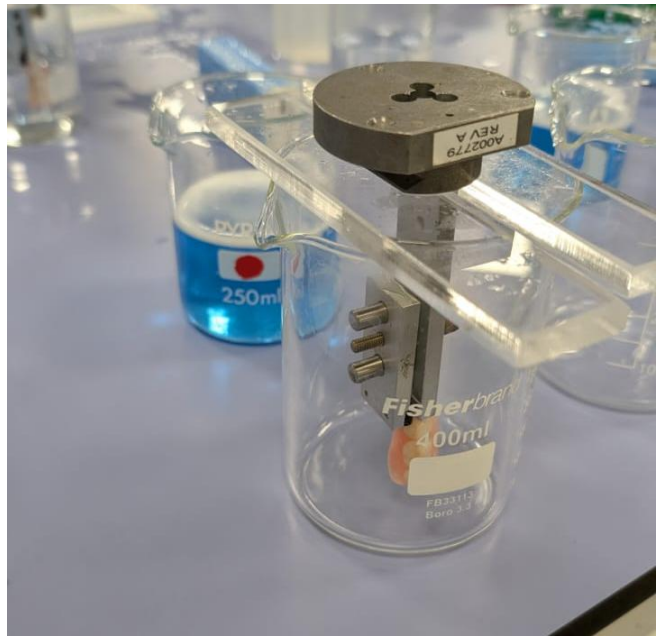
DESIGNED ELECTRIC CAFÉ RACER FOR AMT MOTOCORP 2019 ELECTRIC VEHICLE COMPETITION

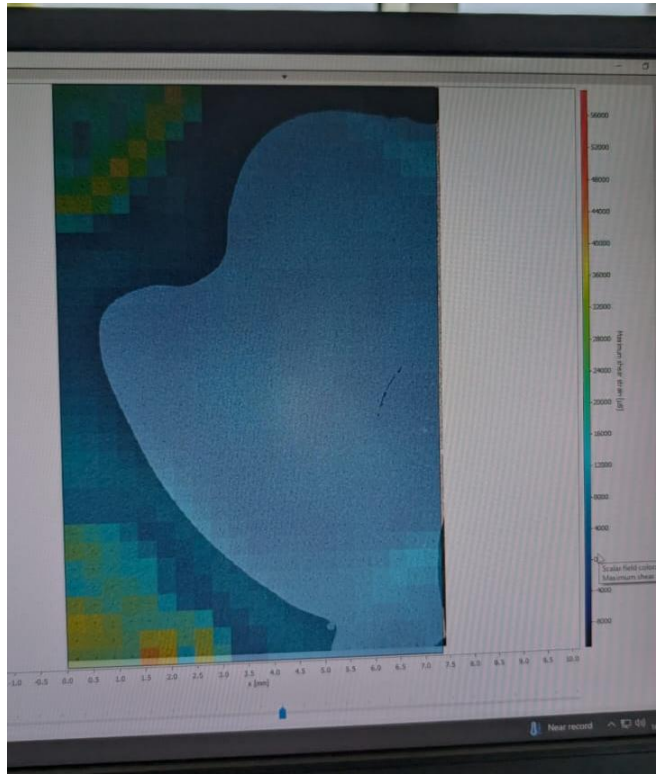


Surface Abrasion Test Project between University of Portsmouth and Haleon



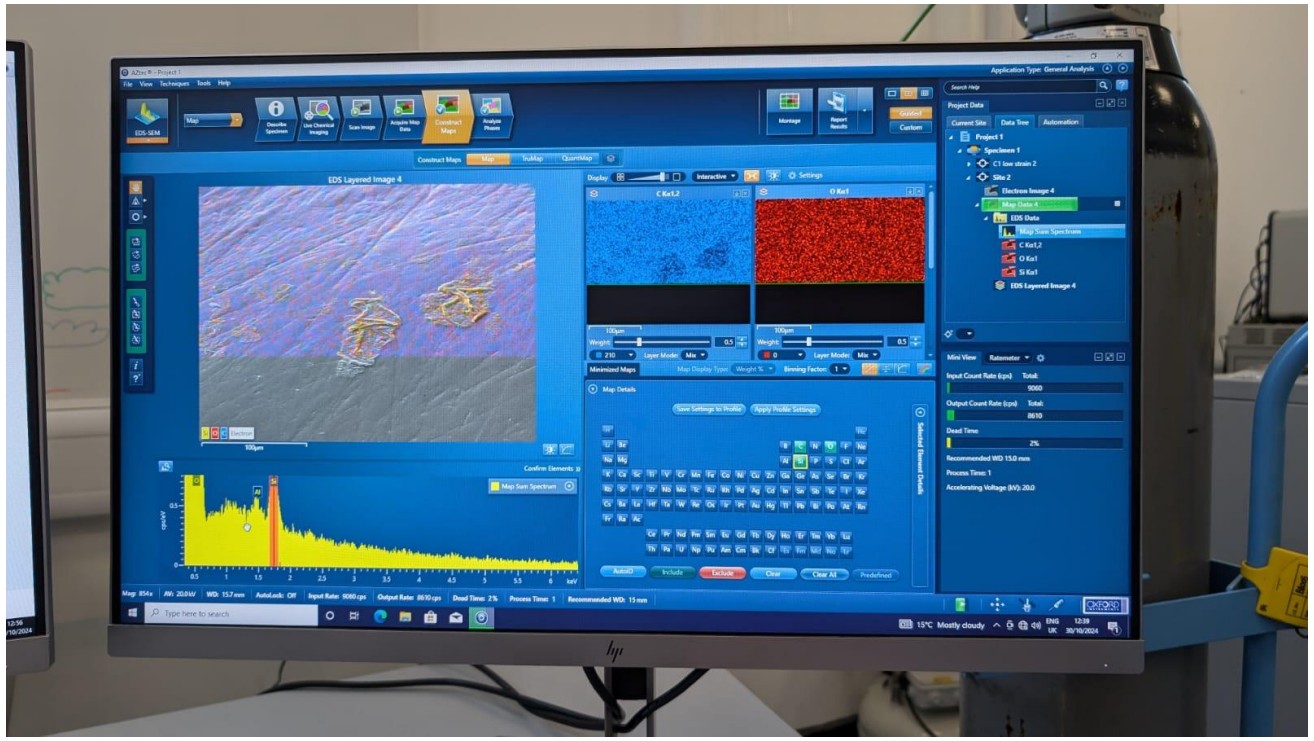
Grew Biofilm on Dental Implant



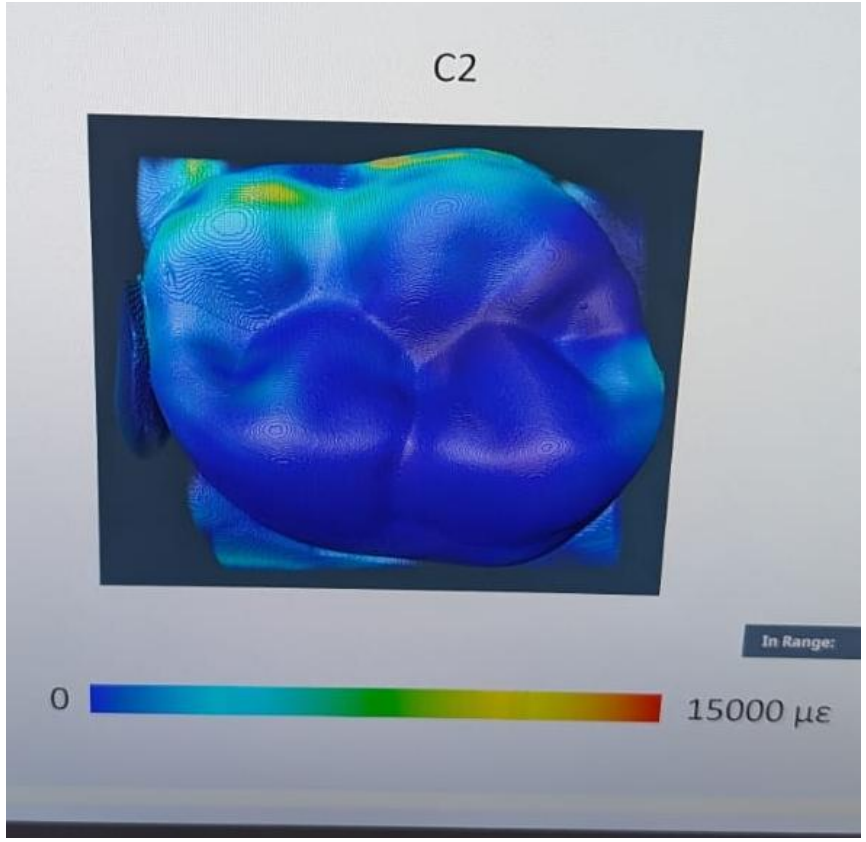
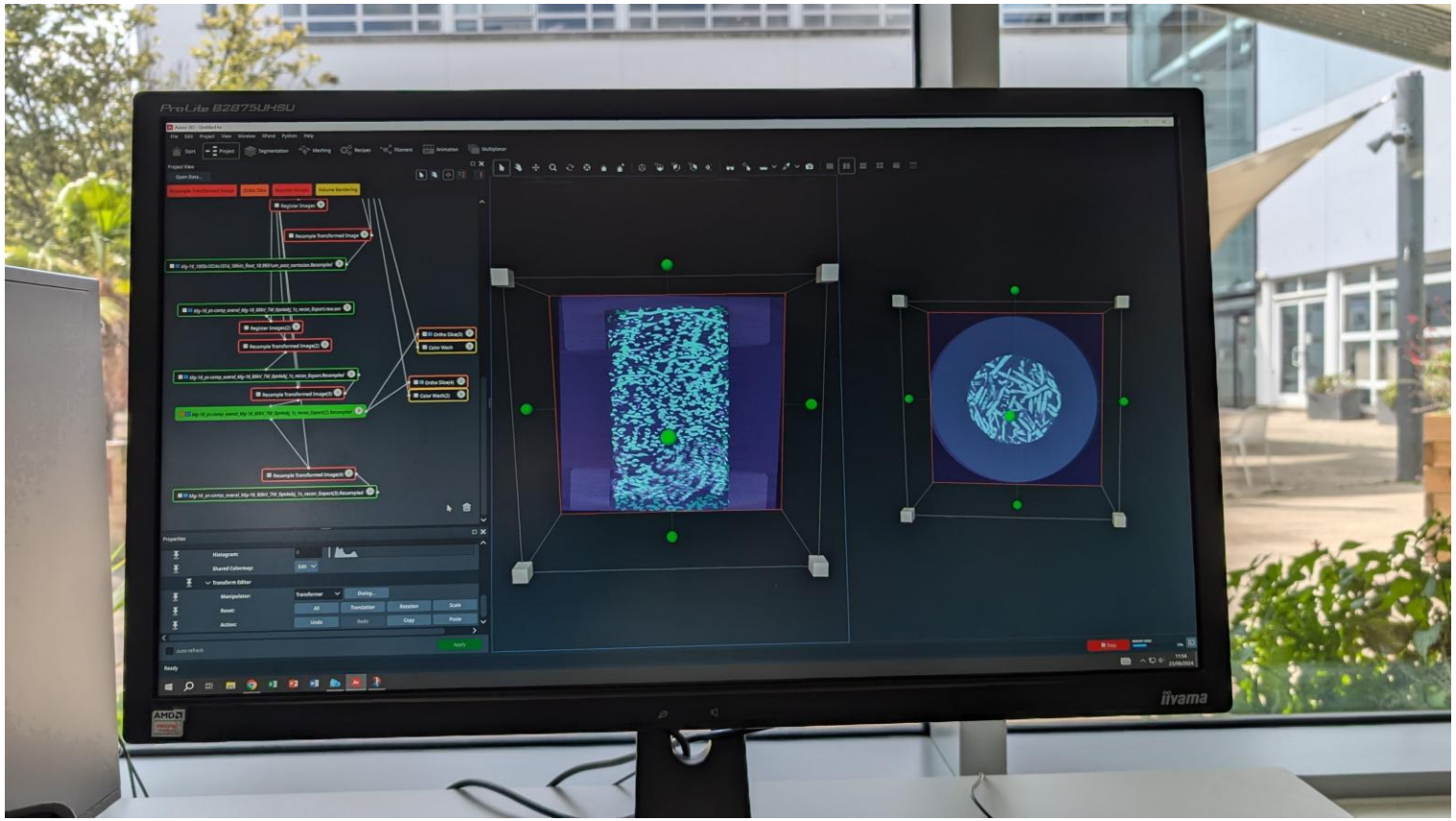


Used Digital Volume Corelation to calculate Strain on Pre vs Post XCT Scans

Correlated abrasion region using SEM



Applied strain map using Avizo



Magnesium implants under compression XCT Scan study for regenerative bone support using AVIZO